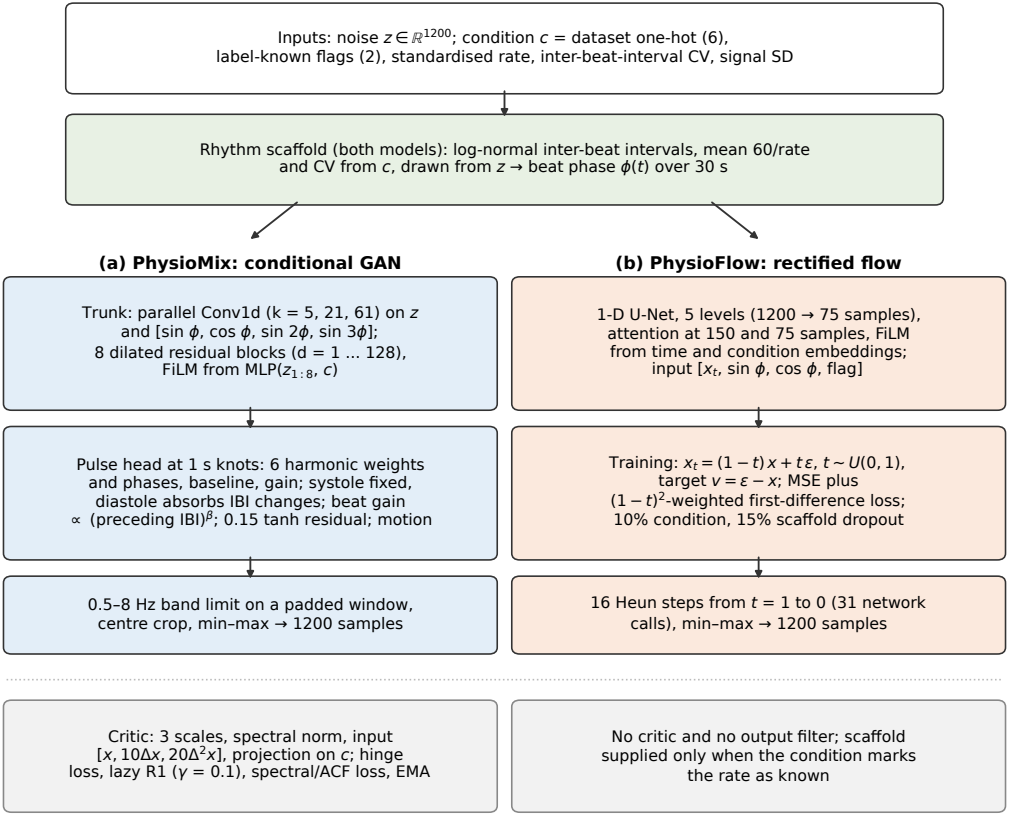
Inputs: noise $z \in \mathbb{R}^{1200}$; condition c = dataset one-hot (6),
label-known flags (2), standardised rate, inter-beat-interval CV, signal SD
Rhythm scaffold (both models): log-normal inter-beat intervals, mean 60/rate
and CV from c, drawn from $z \to$ beat phase $\phi(t)$ over 30 s
(a) PhysioMix: conditional GAN
(b) PhysioFlow: rectified flow
Trunk: parallel Conv1d (k = 5, 21, 61) on z
and [sin ϕ, cos ϕ, sin 2ϕ, sin 3ϕ];
8 dilated residual blocks (d = 1 ... 128),
FiLM from MLP($z_{1:8}$, c)
1-D U-Net, 5 levels (1200 → 75 samples),
attention at 150 and 75 samples, FiLM
from time and condition embeddings;
input [x_t, sin ϕ, cos ϕ, flag]
Pulse head at 1 s knots: 6 harmonic weights
and phases, baseline, gain; systole fixed,
diastole absorbs IBI changes; beat gain
$\propto$ (preceding IBI)$^{\beta}$; 0.15 tanh residual; motion
Training: $x_t = (1-t)x + t\varepsilon$, $t \sim U(0,1)$,
target $v = \varepsilon - x$; MSE plus
$(1-t)^2$-weighted first-difference loss;
10% condition, 15% scaffold dropout
0.5–8 Hz band limit on a padded window,
centre crop, min–max → 1200 samples
16 Heun steps from t = 1 to 0 (31 network
calls), min–max → 1200 samples
Critic: 3 scales, spectral norm, input
$[x, 10\Delta x, 20\Delta^2 x]$, projection on c; hinge
loss, lazy R1 (γ = 0.1), spectral/ACF loss, EMA
No critic and no output filter; scaffold
supplied only when the condition marks
the rate as known